


NOTE:-

- (I) The letter in bold font represents the vector quantity.
- (II) Make necessary assumptions if required.
- (III) Calculator is allowed.
- (IV) All questions carry equal marks until unless stated.

Question 1 (Total Marks 6)

- a) Express following vector in Cartesian form: $\mathbf{A} = \frac{10}{r} \mathbf{a}_r + r \cos \theta \mathbf{a}_\theta + \mathbf{a}_\phi$
- b) Determine the flux of $\mathbf{D} = \rho^2 \cos^2(\varphi) \mathbf{a}_\rho + z \sin(\varphi) \mathbf{a}_\phi$ over the closed surface of the cylinder defined by $0 \leq z \leq 1$, $\rho = 4$. Verify the divergence theorem for this case.

Question 2 (Total Marks 6)

- a) A volume charge with density $\rho_v = 5\rho^2 z \mu\text{C}/\text{m}^3$ exists in a region defined by $0 < \rho < 2$, $0 < z < 1$, and $30^\circ < \varphi < 90^\circ$. Calculate the total charge in the region.
- b) A charge distribution with spherical symmetry has volume charge density

$$\rho_v = \begin{cases} \frac{\rho_0 r}{R}, & 0 \leq r \leq R \\ 0, & r > R \end{cases}$$

Determine the electric field intensity $\mathbf{E}$ everywhere.

Question 3 (Total Marks 6)

- a) Two conduction cones ($\theta = \frac{\pi}{10}$) and ($\theta = \frac{\pi}{6}$) of infinite extent are separated by infinitesimal gap at $r = 0$. If $V(\theta = \frac{\pi}{10}) = 0$ and $V(\theta = \frac{\pi}{6}) = 50 \text{ V}$, find the V and $\mathbf{E}$ between the cones.
- b) Region 1, described by $3x + 4y \geq 10$, is free space, whereas Region 2, described by $3x + 4y \leq 10$, is a magnetic material for which $\mu = 10\mu_0$. Assuming that the boundary between the material and free space is current-free, find $\mathbf{B}_2$ if:

$$\mathbf{B}_1 = 0.1 \mathbf{a}_x + 0.4 \mathbf{a}_y + 0.2 \mathbf{a}_z \text{ Wb/m}^2.$$

Question 4 (Total Marks 6)

- a) Plane $y = 1$ carries current $\mathbf{K} = 50 \mathbf{a}_z \text{ mA/m}$. Find $\mathbf{H}$ at
 - (i) $(0, 0, 0)$
 - (ii) $(1, 5, -3)$
- b) Derive the expression for the magnetic field intensity $\mathbf{H}$ due to an infinite sheet of current placed in the xy -plane. Assume that the surface current flows in the y -direction. Obtain the expression for $\mathbf{H}$ for points both above ($z > 0$) and below ($z < 0$) the current sheet.

Question 5 (Total Marks 6)

- a) Write all the Maxwell's equations of point and integral forms for time varying fields with their physical significances. (2)
- b) In a medium characterized by $\sigma = 0, \mu = \mu_0, \epsilon = 4\epsilon_0$, the electric field is given by $\mathbf{E} = 20 \sin(10^8 t - \beta z) \mathbf{a}_y \text{ V/m}$. (4)
 Calculate:
 (i) the phase constant β , and (ii) the magnetic field intensity $\mathbf{H}$.